Functional Itô Calculus and Path-Dependent Option Greeks

application to an Asian option with floating strike

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Motivation: beyond the Markov property

In mathematical finance, many contracts depend not only on the current asset price, but on the whole trajectory:

- **Asian options** — payoff on time averages;
- **Lookback options** — depending on maxima/minima;
- **Realized volatility** — accumulated variance;
- **Functional volatility** — explicit path dependence.

Denoting the random variable of the process X at time t by x_t and the process up to time t by X_t, that is, $X_t = (x_s)_{0\le s\le t}$, consider the Black–Scholes model (or Geometric Brownian Motion),

$$\mathrm{d}s_t = rs_t\,\mathrm{d}t + \sigma s_t\,\mathrm{d}w_t$$

In it, the price of a European option is a function of time and of the current price only, $V(t, s_t)$. A path-dependent derivative, by contrast, pays based on the entire observed trajectory, so its price is a functional defined on the space of paths, $V(t, X_t)$. The relevant dependence is no longer a single point, but the whole history. It is this change of domain that the functional approach formalizes.

Space of paths and Dupire derivatives

Let $T > 0$ and

$$\Lambda = \bigcup_{t\in[0,T]} \Lambda_t$$

where Λ_t is the set of càdlàg paths on $[0, t]$, with $Y_t \in \Lambda_t$.

Horizontal extension (*flat extension*). For $h > 0$,

$$Y_{t,h}(u) = \begin{cases} Y_t(u), & u \le t \\ Y_t(t), & t < u \le t + h \end{cases}$$

The functional time derivative is

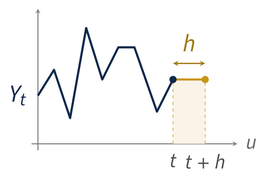
$$\boxed{\Delta_t F(Y_t) = \lim_{h\downarrow 0} \frac{F(Y_{t,h}) - F(Y_t)}{h}}$$

Vertical perturbation (*bump*). For $\varepsilon \in \mathbb{R}$,

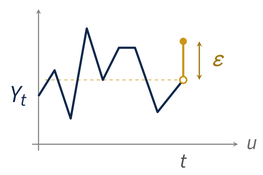
$$Y_t^\varepsilon(u) = \begin{cases} Y_t(u), & u < t \\ Y_t(t) + \varepsilon, & u = t \end{cases}$$

The functional space derivative is

$$\boxed{\Delta_x F(Y_t) = \lim_{\varepsilon\to 0} \frac{F(Y_t^\varepsilon) - F(Y_t)}{\varepsilon}}$$



(a) Horizontal extension



(b) Vertical perturbation

Functional Itô formula

Let X be a continuous semimartingale and $F \in \mathbb{C}^{1,2}$ in the functional sense. The functional Itô formula states that

$$\boxed{\mathrm{d}F(X_t) = \Delta_t F(X_t)\,\mathrm{d}t + \Delta_x F(X_t)\,\mathrm{d}x_t + \frac{1}{2}\Delta_{xx} F(X_t)\,\mathrm{d}\langle x\rangle_t}$$

Compare with the classical formula:

$$\mathrm{d}f(t, x_t) = \partial_t f(t, x_t)\,\mathrm{d}t + \partial_x f(t, x_t)\,\mathrm{d}x_t + \frac{1}{2}\partial_{xx} f(t, x_t)\,\mathrm{d}\langle x\rangle_t$$

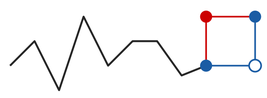
The difference lies in the *domain*: the classical formula acts on functions of finite-dimensional variables; the functional one, on functionals defined on the space of paths.

Path dependence and Lie bracket

Jazaerli and Saporito introduce the bracket

$$\boxed{\mathcal{L}F = [\Delta_t, \Delta_x](F) = \Delta_t\Delta_x F - \Delta_x\Delta_t F}$$

which measures the non-commutativity between time and space perturbations. Geometrically, it compares the operations of vertically perturbing a path and horizontally extending it.



Commutativity diagram

Classification of path dependence

- $\mathcal{L}F = 0$: locally weakly path-dependent functional;
- $\mathcal{L}F \neq 0$: strongly path-dependent functional.

This distinction yields probabilistic representations of the Greeks, with Monte Carlo weights tailored to the dependence structure of the contract.

Financial application: pricing and Greeks

By Girsanov's theorem, we may consider the dynamics under the risk-neutral measure

$$\mathrm{d}s_t = rs_t\,\mathrm{d}t + \sigma(S_t)\,\mathrm{d}w_t$$

For a payoff $g : \Lambda_T \to \mathbb{R}$,

$$V(Y_t) = e^{-r(T-t)}\mathbb{E}\left[g(X_T) \mid Y_t\right] = e^{-r(T-t)}\mathbb{E}\left[g(X_T^{Y_t})\right]$$

where X^{Y_t} coincides with Y_t up to t and evolves according to the model. Under regularity, the functional pricing PDE is

$$\boxed{\Delta_t V(Y_t) + ry_t\Delta_x V(Y_t) + \frac{1}{2}\sigma^2(Y_t)\Delta_{xx} V(Y_t) - rV(Y_t) = 0}$$

The Greeks are functional derivatives of the price:

$$\Delta = \Delta_x V, \qquad \Gamma = \Delta_{xx} V, \qquad \text{Vega} = \partial_\sigma V$$

The goal is to obtain probabilistic representations of these quantities, allowing estimation by Monte Carlo.

Delta representation via weighted expectation

Under suitable conditions,

$$\boxed{\Delta = \mathbb{E}[g(X_T)\,\pi]}$$

where π is a stochastic weight built from the asset dynamics and its tangent process. For functional volatility,

$$\mathrm{d}s_t = \sigma(S_t)\,\mathrm{d}w_t$$

we define the tangent process $\mathrm{d}z_t = \Delta_x\sigma(S_t)\,z_t\,\mathrm{d}w_t$ and $z_0 = 1$ $\left(\text{or } z_t = \frac{\partial s_t}{\partial s_0}\right)$ and obtain:

$$\Delta = \mathbb{E}\left[g(X_T)\,\frac{1}{T-t}\int_t^T \frac{z_s}{\sigma(X_s)}\,\mathrm{d}w_s \,\middle|\, Y_t\right] \xrightarrow{t=0} \Delta = \mathbb{E}\left[g(X_T)\,\frac{1}{T}\int_0^T \frac{z_s}{\sigma(X_s)}\,\mathrm{d}w_s\right]$$

The advantage is to avoid estimating the derivative through perturbations of the initial price.

Numerical experiment I: Asian option with floating strike

A forward-start Asian option with floating strike has payoff

$$\boxed{g(S_T) = \left(s_T - \frac{1}{T-t_1}\int_{t_1}^T s_t\,\mathrm{d}t\right)^+}$$

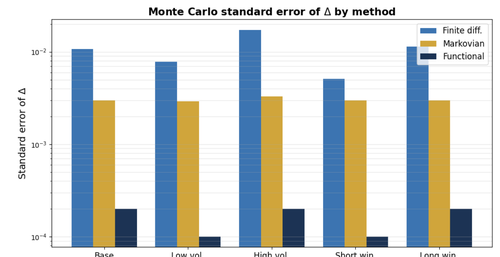
The strike is the arithmetic average of S over $[t_1, T]$. Under Black–Scholes with $r = 0$,

$$\mathrm{d}S_t = \sigma S_t\,\mathrm{d}w_t.$$

For $t < t_1$, the price depends only on S_t, since the averaging window starts at t_1: the path dependence is delayed.

Baseline parameters: $S_0 = 100$, $\sigma = 0.2$, $t_1 = 0.2$, $T = 1$.

	Delta (standard error)		
Scenario	**Finite diff.**	**Markovian**	**Functional**
Baseline ($\sigma = 0.2, t_1 = 0.2$)	0.0432 (0.0107)	0.5398 (0.0030)	0.0411 (0.0002)
Low vol. ($\sigma = 0.15$)	0.0322 (0.0078)	0.5298 (0.0029)	0.0309 (0.0001)
High vol. ($\sigma = 0.30$)	0.0651 (0.0173)	0.5595 (0.0033)	0.0616 (0.0002)
Short window ($t_1 = 0.8$)	0.0170 (0.0051)	0.5398 (0.0030)	0.0204 (0.0001)
Long window ($t_1 = 0.1$)	0.0455 (0.0114)	0.5398 (0.0030)	0.0436 (0.0002)



The functional representation combines the *best of both worlds*: the weight π exploits path dependence to remove the bias of the Markovian benchmark, while avoiding the noise amplification inherent to finite differences. The result is an unbiased Delta estimate with a standard error **50–90× smaller**.

Numerical experiment II: variance knock-out

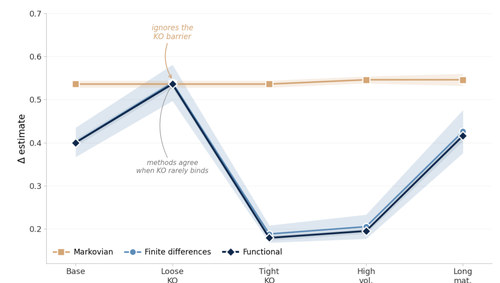
Consider derivatives on the realized variance of the log-price, $QV_t = \langle \log S\rangle_t$. The *VKO Call* (Variance Knock-Out) has payoff

$$\boxed{g(X_T) = (S_T - K)^+\,\mathbf{1}_{\{QV_T < H\}}.}$$

If $QV_T \ge H$, the option is knocked out. Its price depends jointly on S_T and on the volatility realized along the entire path. In simple settings (e.g. local volatility) the price reduces to a function $\psi(t, s, q)$, allowing classical PDEs; in general, however, no such finite-dimensional reduction exists — and the functional approach becomes necessary.

Why is this interesting? The indicator $\mathbf{1}_{\{QV_T<H\}}$ is discontinuous: a small perturbation of the initial price can flip the knock-out event, making finite differences noisy and the Markovian benchmark (which ignores QV_T) severely biased.

	Delta (standard error)		
Scenario	**Finite diff.**	**Markovian**	**Functional**
Base ($H = 0.05$)	0.4010 (0.0166)	0.5364 (0.0040)	0.4003 (0.0036)
Loose KO ($H = 0.20$)	0.5388 (0.0212)	0.5364 (0.0040)	0.5364 (0.0040)
Tight KO ($H = 0.03$)	0.1877 (0.0100)	0.5364 (0.0040)	0.1790 (0.0029)
High vol. ($v_0 = 0.09$)	0.2045 (0.0138)	0.5462 (0.0038)	0.1949 (0.0027)
Long maturity ($T = 2$)	0.4264 (0.0252)	0.5463 (0.0067)	0.4155 (0.0064)



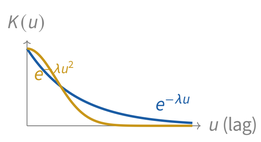
Two complementary effects stand out. **(i) Bias:** the Markovian benchmark is grossly biased — it ignores the knock-out and overestimates Δ by a factor of up to 3× (tight-KO scenario). **(ii) Variance:** finite differences and the functional estimator agree on the value, but the functional estimator cuts the standard error by 3–5×. When the barrier is loose, all three methods collapse — confirming that the discrepancy is driven by path dependence, precisely the regime the functional calculus was designed for.

Beyond options: non-Markovian memory dynamics

A model in which volatility depends on a weighted average of the entire history:

$$\boxed{\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma(Y_t)\,S_t\mathrm{d}w_t, \qquad Y_t = \int_0^t K(t-s)\,S_s\mathrm{d}s,}$$

with kernel $K(u) = e^{-\lambda u^2}$. The price depends on the whole trajectory through Y_t: the classical ansatz $V = V(t, S_t)$ is insufficient.



Tractable case. For an *exponential* kernel $K(u) = e^{-\lambda u}$, a single extra state closes the dynamics:

$$\mathrm{d}Y_t = (S_t - \lambda Y_t)\,\mathrm{d}t,$$

and the price reduces to a classical PDE in (t, S_t, Y_t).

This case. For the *Gaussian* kernel $e^{-\lambda u^2}$, no finite vector of statistics suffices: the kernel has infinite memory rank, and state augmentation is impossible.

When memory cannot be compressed into finitely many state variables, the functional calculus is not an alternative but a *necessity*: it works directly on the space of paths, $V = V\big(t, (S_s)_{s\le t}\big)$, and brings the historical memory into the pricing equation itself.

Outlook

- **Geometry.** Càdlàg paths as points of an infinite-dimensional manifold; Δ_t, Δ_x as vector fields and $\mathcal{L}F$ as their Lie bracket.
- **Functional analysis.** Adjoints Δ_t^*, Δ_x^*, Δ_{xx}^*, integration by parts, weak and Galerkin formulations on path space.
- **Beyond finance.** Non-Markovian Langevin dynamics (physics), sequential and functional data analysis (statistics), history-dependent epidemic models (epidemiology).

References

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